

Binary Change Detection Report

1. Abstract

This work addresses the problem of binary change detection using co-registered Electro-Optical (EO) and Synthetic Aperture Radar (SAR) image pairs. The objective is to classify every pixel as either **Change** or **No-Change** between pre-event and post-event observations. EO-SAR change detection is a challenging remote sensing problem because EO and SAR imagery possess fundamentally different physical characteristics. EO imagery captures spectral and texture information, while SAR imagery captures microwave backscatter and is highly affected by speckle noise.

After analysing the dataset characteristics and task constraints specified in the assignment, a **Siamese UNet architecture with a ResNet-34 encoder** was selected. The network was trained using a combination of **Weighted Binary Cross Entropy (BCE)** and **Dice Loss** to mitigate severe class imbalance. Experimental analysis showed strong recall on validation data but weak generalization on unseen test regions due to domain shift and modality heterogeneity.

The report also evaluates architectural trade-offs, dataset limitations, and future directions including transformer-based fusion and domain adaptation methods.

2. Problem Understanding

The assignment requires binary segmentation of damaged regions using EO-SAR image pairs. The dataset originally contains four semantic labels:

Original Label	Meaning	Remapped Label
0	Background	0
1	Intact	0
2	Damaged	1
3	Destroyed	1

Thus, the task becomes a **binary semantic segmentation problem**:

- **0** → **No Change**
- **1** → **Change**

The model must predict a dense binary mask for each image pair.

The assignment additionally emphasizes:

- Dataset understanding before modelling
- Handling severe class imbalance
- Justification of architectural choices

- Evaluation using:
 - IoU
 - Precision
 - Recall
 - F1 Score
 - Confusion Matrix

3. Dataset Analysis

3.1 EO and SAR Modality Characteristics

The dataset contains:

- **Pre-event EO imagery**
 - Multi-channel optical imagery
 - Rich texture and structural information
 - Sensitive to illumination and weather
- **Post-event SAR imagery**
 - Single-channel radar backscatter
 - Robust to cloud cover and illumination
 - Contains speckle noise
 - Different statistical distribution compared to EO

This heterogeneous modality setup makes direct feature comparison difficult.

3.2 Class Imbalance

A major challenge observed in the dataset is severe imbalance between changed and unchanged pixels.

Approximate distribution:

Class	Relative Frequency
No-Change	~98%
Change	~2%

This imbalance strongly biases training toward predicting the majority class. Without mitigation strategies, the network would achieve deceptively high accuracy while failing to detect damaged regions.

3.3 Dataset Challenges

Several practical remote sensing challenges were identified:

a) Domain Shift

Different disaster events contain:

- Different building styles
- Different vegetation distributions
- Different terrain textures
- Different SAR noise patterns

This causes poor generalization between train and test regions.

b) SAR Speckle Noise

SAR imagery contains multiplicative noise patterns that resemble structural damage.

c) Small Target Structures

Destroyed buildings often occupy very small spatial regions, making segmentation difficult.

d) Modality Gap

EO and SAR encode different physical properties, making feature alignment non-trivial.

4. Literature Survey

Change detection in remote sensing has evolved significantly from traditional image differencing approaches to deep learning-based semantic segmentation frameworks. Early methods such as Change Vector Analysis (CVA), Principal Component Analysis (PCA), and thresholding techniques were computationally simple but struggled under illumination variations, SAR speckle noise, and heterogeneous modality differences [1].

Recent advances in deep learning have substantially improved performance in change detection tasks. Fully Convolutional Siamese Networks (FC-Siam) introduced by Daudt et al. demonstrated that shared-weight encoders effectively learn temporal feature differences while reducing parameter complexity [2].

UNet-based architectures remain highly influential in segmentation because encoder-decoder structures combined with skip connections preserve spatial detail while extracting high-level semantic representations [3].

More recently, transformer-based models such as ChangeFormer introduced hierarchical self-attention mechanisms for long-range dependency modelling in change detection tasks [4]. These architectures outperform conventional CNNs in large-scale scene understanding but require substantially higher computational resources.

For EO-SAR heterogeneous fusion, modality alignment remains challenging because optical imagery captures spectral reflectance while SAR imagery captures microwave backscatter. Several studies explored shared latent feature spaces and cross-modal attention mechanisms to bridge this representation gap [5].

Additionally, attention-based multi-scale architectures have shown strong performance in remote sensing segmentation because they improve contextual reasoning and small object detection [6].

5. Methodology

The proposed framework uses a Siamese UNet architecture with ResNet-34 encoders. UNet was selected because of its effectiveness in dense segmentation tasks requiring precise localization [3].

Shared-weight Siamese encoders were used to learn modality-invariant representations between EO and SAR image pairs, following the design philosophy proposed in FC-Siam networks [2].

The network processes:

- EO pre-event imagery through a 3-channel encoder
- SAR post-event imagery through a modified single-channel encoder

Feature fusion is performed using:

$$|F_{\{EO\}} - F_{\{SAR\}}|$$

where:

- $(F_{\{EO\}})$ = EO feature representation
- $(F_{\{SAR\}})$ = SAR feature representation

This operation explicitly highlights structural changes between temporal observations.

To address severe class imbalance, the loss function combines Weighted Binary Cross Entropy with Dice Loss. Dice-based optimization has shown strong performance in highly imbalanced segmentation problems because it directly optimizes overlap between predicted and ground-truth masks [7].

The final optimization objective is:

$$\text{Loss} = 0.5(\text{BCE}) + 0.5(\text{Dice})$$

Data augmentation techniques were implemented using the Albumentations framework, which provides computationally efficient augmentation pipelines for computer vision tasks [8].

Optimization settings:

- Optimizer: AdamW
- Learning Rate: (1×10^{-4})
- Batch Size: 8
- Scheduler: Cosine Annealing

6. Results and Analysis

The model achieved strong recall performance on validation data, indicating that it successfully learned to identify damaged regions despite severe imbalance.

Metric	Validation	Test
IoU	24.74%	3.50%
Precision	26.45%	4.19%
Recall	79.29%	17.37%
F1 Score	39.67%	6.75%

The large performance drop between validation and test data suggests significant domain shift between training and unseen disaster regions. Similar generalization challenges are commonly reported in remote sensing change detection literature [4][5].

False positives were primarily caused by:

- SAR speckle noise
- Seasonal vegetation changes
- Structural texture variations

7. Error Analysis

7.1 Validation Behaviour

The model achieved relatively high recall on validation data:
Recall = 79.29%

This indicates that the model successfully learned to identify most damaged regions.

However, precision remained relatively low:
Precision = 26.45%

This suggests that the model generated a large number of false positives.

7.2 Test Performance Drop

The model performance decreased dramatically on the unseen test split.

Metric	Validation	Test
IoU	24.74%	3.50%
F1 Score	39.67%	6.75%

This large gap strongly indicates severe domain shift between training and testing distributions [5].

Possible reasons include:

- Different SAR acquisition characteristics
- Different geographic regions
- Different structural layouts
- Seasonal variations

8.Time and Resource Log

Time Breakdown

Activity	Time Spent
Dataset Exploration	2 hours
Literature Review	2 hours
Implementation	4 hours
Training & Debugging	3 hours
Evaluation & Reporting	1 hours

Total Time:
12 hours

Training Hardware

Component	Details
Platform	Google Colab
GPU	NVIDIA T4
VRAM	16 GB
Batch Size	8
Epoch Time	~60 seconds
Total Training Time	~15 minutes

Resource Constraints

Limited GPU memory restricted experimentation with:

- Larger transformer architectures
- Higher-resolution training
- Multi-scale ensembles

Therefore, a computationally efficient Siamese UNet architecture was selected as a practical trade-off between:

- Accuracy
- Memory usage
- Training speed

9. Future Work

Future improvements could include transformer-based architectures such as ChangeFormer, which use hierarchical attention mechanisms for improved contextual modelling in remote sensing change detection [4].

Additionally, adversarial domain adaptation methods could reduce train-test distribution mismatch, which remains a major limitation in disaster-response datasets [5].

Further improvements may include:

- SAR despeckling modules
- Multi-scale attention fusion
- Semi-supervised pseudo-labeling
- Focal Loss optimization for hard examples [9]

10. Conclusion

This project implemented a complete EO-SAR binary change detection pipeline using a Siamese UNet architecture with weighted loss optimization. The approach successfully learned meaningful structural differences despite severe class imbalance and heterogeneous sensor modalities.

The primary limitation observed was weak generalization to unseen geographic regions, highlighting the importance of:

- Robust feature learning
- Domain adaptation
- Transformer-based contextual modelling

Despite these limitations, the implementation remains:

- Reproducible
- Computationally efficient
- Technically justified
- Extensible for future research

11. References

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